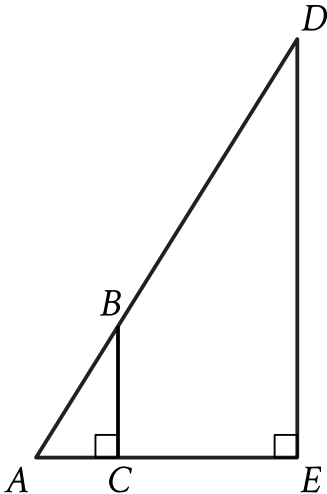


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, $AB = \sqrt{34}$ units, $AC = 3$ units, and $CE = 21$ units. What is the area, in square units, of triangle ADE ?

Correct Answer: 480

Rationale

The correct answer is **480**. It's given in the figure that angle ACB and angle AED are right angles. It follows that angle ACB is congruent to angle AED . It's also given that angle BAC and angle DAE are the same angle. It follows that angle BAC is congruent to angle DAE . Since triangles ABC and ADE have two pairs of congruent angles, the triangles are similar. Sides AB and AC in triangle ABC correspond to sides AD and AE , respectively, in triangle ADE . Corresponding sides in similar triangles are proportional. Therefore, $\frac{AD}{AB} = \frac{AE}{AC}$. It's given that $AC = 3$ units and $CE = 21$ units. Therefore, $AE = 24$ units. It's also given that $AB = \sqrt{34}$ units. Substituting **3** for AC , **24** for AE , and $\sqrt{34}$ for AB in the equation $\frac{AD}{AB} = \frac{AE}{AC}$ yields $\frac{AD}{\sqrt{34}} = \frac{24}{3}$, or $\frac{AD}{\sqrt{34}} = 8$. Multiplying each side of this equation by $\sqrt{34}$ yields $AD = 8\sqrt{34}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Since triangle ADE is a right triangle, it follows that AD represents the length of the hypotenuse, c , and DE and AE represent the lengths of the legs, a and b . Substituting **24** for b and $8\sqrt{34}$ for c in the equation $a^2 + b^2 = c^2$ yields $a^2 + (24)^2 = (8\sqrt{34})^2$, which is equivalent to $a^2 + 576 = 64(34)$, or $a^2 + 576 = 2,176$. Subtracting **576** from both sides of this equation yields $a^2 = 1,600$. Taking the square root of both sides of this equation yields $a = \pm 40$. Since a represents a length, which must be positive, the value of a is **40**. Therefore, $DE = 40$. Since DE and AE represent the lengths of the legs of triangle ADE , it follows that DE and AE can be used to calculate the area, in square units, of the triangle as $\frac{1}{2}(40)(24)$, or **480**. Therefore, the area, in square units, of triangle ADE is **480**.